

Notes Power Model D-MIMO

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I. CENTRAL MIMO POWER MODEL

We adopt the parametric power model for upper mid-band (FR3) base stations of [1]. The base station (BS) is equipped with M_{ant} antennas, an equal number of power amplifiers (PAs, used in the downlink) and low-noise amplifiers (LNAs, used in the uplink), and M_{RF} radio-frequency (RF) chains. It serves K single-antenna users over a bandwidth B at carrier frequency f_c using OFDM and space-division multiple access. Setting $M_{\text{RF}} = M_{\text{ant}}$ selects fully-digital beamforming; $M_{\text{RF}} < M_{\text{ant}}$ selects hybrid partially-connected beamforming with $M_{\text{ant}}/M_{\text{RF}}$ phase shifters per RF chain.

A. Model structure

The total consumed power is split into three hardware blocks; digital signal processing, analog signal processing, and the PAs, each divided by its supply-and-cooling efficiency $\eta_{c,\text{sc}} \in (0, 1]$:

$$P_{\text{cons}} = \frac{\overline{P_{\text{dig}}}}{\eta_{\text{dig,sc}}} + \frac{\overline{P_{\text{ana}}}}{\eta_{\text{ana,sc}}} + \frac{\overline{P_{\text{PA}}}}{\eta_{\text{PA,sc}}}. \quad (1)$$

Each block is built from the working-mode (active) consumption of its subcomponents, averaged over the operating modes within a frame. We first define this frame averaging (Section I-B) and then reuse it as a function for every component (Sections I-C to I-E).

B. Frame structure and operating-mode averaging

A frame of duration T_f contains a downlink (DL) and an uplink (UL) phase under time division duplexing (TDD), with duration ratios $\tau_{\text{DL}}, \tau_{\text{UL}} \in [0, 1]$ and $\tau_{\text{UL}} = 1 - \tau_{\text{DL}}$. Within each direction $i \in \{\text{DL}, \text{UL}\}$ the time is further subdivided into three operating modes, plus an idle mode when the direction is inactive:

- *reference signalling*, for a time fraction $\tau_i \tau_{i,\text{sig}}$;
- *data transmission/reception*, for a fraction $\tau_i (1 - \tau_{i,\text{sig}}) \frac{N_{a,i}}{N_i}$;
- *micro-sleep* (DTX), for a fraction $\tau_i (1 - \tau_{i,\text{sig}}) \left(1 - \frac{N_{a,i}}{N_i}\right)$;
- *idle*, for a fraction $(1 - \tau_i)$,

where N_i is the number of OFDM symbols (time slots) of direction i and $N_{a,i} \leq N_i$ the subset that are *active* (carrying data). The *active-resource time ratio* $N_{a,i}/N_i$ is what sets the data-versus-micro-sleep split: a slot is either active or not, regardless of how many users share it. with $\tau_{i,\text{sig}} \in [0, 1]$ the signalling fraction of direction i .

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a) *Average physical resource load*: The data-mode weight is more conveniently expressed through the *average physical resource load* $\bar{x}_i$, which we now relate to the time ratio $N_{a,i}/N_i$. Let K_{max} be the maximum number of users that can be served (the number of spatial streams) and Q_{max} the number of available data subcarriers per active OFDM symbol. The instantaneous load of user k in time slot n is the ratio of utilised to available data subcarriers,

$$\ell_i[k, n] = \frac{Q_i[k, n]}{Q_{\text{max}}}, \quad i \in \{\text{DL}, \text{UL}\}, \quad (2)$$

for $k = 1, \dots, K_{\text{max}}$ and $n = 1, \dots, N_i$. Averaging over all K_{max} potential users and N_i slots gives the average physical resource load

$$\bar{x}_i \triangleq \bar{\ell}_i = \frac{1}{N_i} \sum_{n=1}^{N_i} \left(\frac{1}{K_{\text{max}}} \sum_{k=1}^{K_{\text{max}}} \frac{Q_i[k, n]}{Q_{\text{max}}} \right). \quad (3)$$

Under space-division multiple access (no OFDMA), an active user in an active slot is allocated *all* data subcarriers, $Q_i[k, n] = Q_{\text{max}}$. Indexing the K_i active users as $\{1, \dots, K_i\}$ and the $N_{a,i}$ active slots as $\{1, \dots, N_{a,i}\}$, each active-user/active-slot pair contributes 1 to (3) and every other pair 0; the double sum equals $K_i N_{a,i}$ and

$$\bar{x}_i = \frac{N_{a,i}}{N_i} \frac{K_i}{K_{\text{max}}}. \quad (4)$$

b) *Equivalence and modelling assumption*: The duty-cycle split in the frame structure is governed by the active-resource time ratio $N_{a,i}/N_i$, whereas the power model uses the average load $\bar{x}_i$ as the data-mode weight. Comparing (4) with the data fraction $\tau_i (1 - \tau_{i,\text{sig}}) N_{a,i}/N_i$, the two agree,

$$\bar{x}_i = \frac{N_{a,i}}{N_i} \iff \frac{K_i}{K_{\text{max}}} = 1 \iff K_i = K_{\text{max}}, \quad (5)$$

i.e. only when every spatial stream is scheduled. When the maximum number of users is active the average load is simply the ratio of active to total slots; for $K_i < K_{\text{max}}$ the average load (4) is the true active fraction scaled by $K_i/K_{\text{max}} < 1$. The number of active users K_i separately sets the transmit power and the achievable rate, but not whether a slot is active. We use $\bar{x}_i$ as the data-mode weight throughout (and $1 - \bar{x}_i$ as the micro-sleep/DTX fraction); this is therefore *exact* in the full-spatial-load regime $K_i = K_{\text{max}}$ evaluated here and an approximation otherwise.

In micro-sleep and idle a component drops to a fraction of its working consumption, set by the reduction factors $\delta_c^\mu, \delta_c^0 \in [0, 1]$. For a component c with working power P^a during data, P^s during signalling, and base power P^0 scaled in the reduced

modes, the frame-averaged consumption of direction i is the time-weighted sum

$$\begin{aligned} \mathcal{A}_i^c(P^a, P^s, P^0) &\triangleq \bar{x}_i \tau_i (1 - \tau_{i,\text{sig}}) P^a + \tau_i \tau_{i,\text{sig}} P^s \\ &\quad + (1 - \bar{x}_i) \tau_i (1 - \tau_{i,\text{sig}}) \delta_c^\mu P^0 \\ &\quad + (1 - \tau_i) \delta_c^0 P^0. \end{aligned} \quad (6)$$

For the digital and analog blocks the consumption is identical in all working modes, $P^a = P^s = P^0 = P$, and we use the shorthand $\mathcal{A}_i^c(P) \triangleq \mathcal{A}_i^c(P, P, P)$. Only the PA distinguishes the three levels (Section I-C).

Each direction's hardware is averaged over the *whole* frame, and the two directions are accounted for separately, e.g. for the digital block $\bar{P}_{\text{dig}} = \mathcal{A}_{\text{DL}}^{\text{dig}}(P_{\text{dig},\text{DL}}) + \mathcal{A}_{\text{UL}}^{\text{dig}}(P_{\text{dig},\text{UL}})$. The idle term $(1 - \tau_i) \delta_c^0 P^0$ is therefore the reduced consumption of direction i 's chain while that direction is *not* active, i.e. while the opposite direction occupies the frame. For $i = \text{DL}$ this fraction equals τ_{UL} only because the gap-free TDD frame satisfies $\tau_{\text{UL}} = 1 - \tau_{\text{DL}}$; it represents the idle DL chain, not the active UL processing, which is captured by the separate $\mathcal{A}_{\text{UL}}$ term.

Remark 1 (Load-dependent split). *Only the data term in (6) scales with the load $\bar{x}_i$, while the micro-sleep term scales with $(1 - \bar{x}_i)$. Collecting the terms proportional to $\bar{x}_i$ separates each averaged consumption into a load-dependent part and a load-independent part, which is used in the analysis to attribute consumption to the delivered traffic.*

C. Power amplifiers

The PAs transmit only in the DL. With a per-PA output power $P_a = P_{T,\text{DL}}/M_{\text{ant}}$, where $P_{T,\text{DL}}$ is the total DL transmit power, the instantaneous consumption of one PA is modelled as

$$P_{\text{PA}}(p) = \frac{1}{\eta_{\text{PA},\text{max}}} [\xi P_{\text{max}} + (1 - \xi) P_{\text{max}}^{1-\alpha} p^\alpha], \quad (7)$$

where p is the output power, P_{max} the maximum output power (at a fixed back-off from saturation), $\eta_{\text{PA},\text{max}}$ the maximum PA efficiency, $\xi \in [0, 1]$ the static-to-dynamic consumption ratio, and $\alpha \in [0, 1]$ the efficiency exponent. Averaging over the DL frame and summing over the M_{ant} PAs gives

$$\bar{P}_{\text{PA}} = M_{\text{ant}} \mathcal{A}_{\text{DL}}^{\text{PA}}(P_{\text{PA}}(P_a), P_{\text{PA}}(\zeta_{\text{sig}} P_{\text{max}}), P_{\text{PA}}(0)), \quad (8)$$

where $\zeta_{\text{sig}} \in (0, 1]$ is the fraction of P_{max} used during signalling, and the zero-output consumption $P_{\text{PA}}(0)$ is the base level scaled in micro-sleep and idle.

D. Analog signal processing

The analog block comprises, per RF chain, the DACs (DL) or ADCs (UL), RF filters, mixers, M_{PS} phase shifters, and the LNAs (UL); the local oscillator (LO) is shared across chains. With $M_{\text{PS}} = M_{\text{ant}}/M_{\text{RF}}$ for hybrid beamforming ($M_{\text{PS}} = 0$

for fully-digital), the working-mode consumption per direction is

$$P_{\text{ana},\text{DL}} = P_{\text{LO}} + M_{\text{RF}}(2P_{\text{DAC}} + 2P_{\text{filt},\text{RF}} + 2P_{\text{mix}} + M_{\text{PS}}P_{\text{PS}}), \quad (9)$$

$$P_{\text{ana},\text{UL}} = P_{\text{LO}} + M_{\text{RF}}(2P_{\text{ADC}} + 2P_{\text{filt},\text{RF}} + 2P_{\text{mix}} + M_{\text{PS}}P_{\text{PS}} + \frac{M_{\text{ant}}}{M_{\text{RF}}}P_{\text{LNA}}), \quad (10)$$

with the factor 2 accounting for the in-phase and quadrature branches. The subcomponents are modelled as

$$P_{\text{DAC}} = \Xi_{\text{DAC},1} 2^{b_{\text{DAC}}} + \Xi_{\text{DAC},2} b_{\text{DAC}} f_{\text{DAC}}, \quad (11)$$

$$P_{\text{ADC}} = W_{\text{ADC}} f_{\text{ADC}} 2^{b_{\text{ADC}}}, \quad (12)$$

$$P_{\text{mix}} = \Xi_{\text{mix}} f_c, \quad (13)$$

$$P_{\text{PS}} = \Xi_{\text{PS}} B, \quad (14)$$

$$P_{\text{LNA}} = \Xi_{\text{LNA}} B, \quad (15)$$

where $b_{\text{DAC}}, b_{\text{ADC}}$ are the effective resolutions, $f_{\text{DAC}}, f_{\text{ADC}}$ the converter sampling rates, W_{ADC} the ADC Walden figure-of-merit, and $\Xi_\bullet$ the respective figures-of-merit. The frame-averaged analog consumption is

$$\bar{P}_{\text{ana}} = \mathcal{A}_{\text{DL}}^{\text{ana}}(P_{\text{ana},\text{DL}}) + \mathcal{A}_{\text{UL}}^{\text{ana}}(P_{\text{ana},\text{UL}}) + P_{\text{ana},\text{misc}}. \quad (16)$$

E. Digital signal processing

In the DL the digital block runs the channel encoder, MIMO precoder, IFFT, DPD, and baseband (BB) filter; in the UL the channel decoder, MIMO combiner, FFT, and BB filter. Each subcomponent consumption is a static (signal-independent) term plus a dynamic term equal to its complex giga-operations-per-second (GOPS) Ξ divided by the computational efficiency η (in GOPS/W) of the chosen FPGA/ASIC implementation:

$$P_{\text{enc}} = P_{\text{enc},s} + \frac{f_{s,\text{I}}}{\eta_{\text{enc}}} \frac{14}{24} \frac{R_{\text{DL}}}{\tilde{B}}, \quad (17)$$

$$P_{\text{dec}} = P_{\text{dec},s} + \frac{f_{s,\text{I}}}{\eta_{\text{dec}}} \frac{175}{6} \frac{R_{\text{UL}}}{\tilde{B}}, \quad (18)$$

$$P_{\text{pre}} = \bar{M}_{\text{RF}} P_s + \frac{f_{s,\text{I}}}{\eta_{\text{pre}}} M_{\text{RF}} \Xi_{\text{pre}}, \quad (19)$$

$$P_{\text{comb}} = \bar{M}_{\text{RF}} P_s + \frac{f_{s,\text{I}}}{\eta_{\text{comb}}} M_{\text{RF}} 2K, \quad (20)$$

$$P_{\text{IFFT}} = P_{\text{IFFT},s} + \frac{f_{s,\text{II}}}{\eta_{\text{IFFT}}} M_{\text{RF}} \frac{3}{2} \log_2 Q_{\text{IFFT}}, \quad (21)$$

$$P_{\text{DPD}} = P_{\text{DPD},s} + \frac{f_{s,\text{II}}}{\eta_{\text{DPD}}} M_{\text{RF}} \Xi_{\text{DPD}}, \quad (22)$$

$$P_{\text{filt},\text{BB}} = \bar{M}_{\text{RF}} P_s + \frac{f_{s,\text{II}}}{\eta_{\text{BB}}} M_{\text{RF}} \Xi_{\text{BB}}, \quad (23)$$

where $R_{\text{DL}}, R_{\text{UL}}$ are the delivered DL/UL sum rates, $\tilde{B}$ the effective bandwidth, $\bar{M}_{\text{RF}} = \lceil M_{\text{RF}}/32 \rceil$ the number of FPGAs (one per 32 chains, sharing the precoder/combiner/BB-filter static power P_s), and Q_{IFFT} the IFFT/FFT size. The two sampling rates are the constellation rate $f_{s,\text{I}} = \mu B$ (for encoding/decoding and MIMO processing) and the oversampled rate $f_{s,\text{II}} = Q_{\text{IFFT}} \Delta f$ (for the OFDM, DPD, and BB operations), with $\mu \in [0, 1]$ and subcarrier spacing Δf . For the zero-forcing

MIMO processor exploiting channel reciprocity, the precoder GOPS is

$$\Xi_{\text{pre}} = 2K + \frac{1}{v_{\text{coh}}} \left(\frac{K^3}{3M_{\text{RF}}} + 3K^2 + K \right), \quad (24)$$

with v_{coh} the number of samples per channel coherence block over which the precoder is reused; the combiner only applies the precoding matrix, giving $\Xi_{\text{comb}} = 2K$. The active digital consumption per direction is the sum of its subcomponents,

$$P_{\text{dig,DL}} = P_{\text{enc}} + P_{\text{pre}} + P_{\text{IFFT}} + P_{\text{DPD}} + P_{\text{filt,BB}}, \quad (25)$$

$$P_{\text{dig,UL}} = P_{\text{dec}} + P_{\text{comb}} + P_{\text{IFFT}} + P_{\text{filt,BB}}, \quad (26)$$

and the frame-averaged digital consumption is

$$\overline{P}_{\text{dig}} = \mathcal{A}_{\text{DL}}^{\text{dig}}(P_{\text{dig,DL}}) + \mathcal{A}_{\text{UL}}^{\text{dig}}(P_{\text{dig,UL}}). \quad (27)$$

F. Delivered rate and energy efficiency

The achievable ergodic sum rate of direction i , which drives the encoder/decoder consumption in (17), follows from the per-user SINRs under zero-forcing beamforming as

$$R_i = \bar{x}_i \tau_i (1 - \tau_{i,\text{sig}}) \frac{\tilde{B}}{q_i} \mathbb{E} \left\{ \sum_{\nu=1}^{q_i} \sum_{k=1}^K \log_2(1 + \text{SINR}_{i,k}[\nu]) \right\}, \quad (28)$$

where the pre-log collects the load, duration, signalling-overhead, and effective-bandwidth-per-subcarrier factors. The base-station energy efficiency then follows as the delivered rate per consumed watt,

$$\text{EE} = \frac{R_{\text{DL}} + R_{\text{UL}}}{P_{\text{cons}}} \quad [\text{bit/J}]. \quad (29)$$

II. DISTRIBUTED MASSIVE MIMO SYSTEM MODEL

A. Downlink System Model

We consider the downlink of a distributed (cell-free) massive multiple input multiple output (MIMO) network in which L access points (APs), each equipped with M antennas, jointly serve K single-antenna users over the same time-frequency resources. The APs are connected through fronthaul links to a central processing unit (CPU) that encodes the data and computes the precoders, so that the LM antennas act as a single distributed array. Transmission uses OFDM with Q data subcarriers, indexed by $q = 1, \dots, Q$; the model below is stated per subcarrier and the same processing is applied on each of them.

Let $\mathbf{h}_{kl}[q] \in \mathbb{C}^M$ denote the channel from AP l to user k on subcarrier q . Stacking the contributions of all APs gives the collective channel

$$\mathbf{h}_k[q] = \begin{bmatrix} \mathbf{h}_{k1}[q] \\ \vdots \\ \mathbf{h}_{kL}[q] \end{bmatrix} \in \mathbb{C}^{LM}, \quad (30)$$

which we model as correlated Rayleigh fading, $\mathbf{h}_k[q] \sim \mathcal{CN}(\mathbf{0}_{LM}, \mathbf{R}_k[q])$. Because the antennas are geographically distributed, the spatial correlation matrix is block-diagonal, $\mathbf{R}_k[q] = \text{diag}(\mathbf{R}_{k1}[q], \dots, \mathbf{R}_{kL}[q]) \in \mathbb{C}^{LM \times LM}$, with each block $\mathbf{R}_{kl}[q] \in \mathbb{C}^{M \times M}$ describing the correlation across the

antennas of AP l ; its scaling captures the large-scale fading between AP l and user k .

The CPU maps the data symbol $s_k[q] \in \mathbb{C}$ of user k , with unit power $\mathbb{E}(|s_k[q]|^2) = 1$ and mutually independent across users, onto the M antennas of each AP. AP l transmits

$$\mathbf{x}_l[q] = \sum_{k=1}^K \mathbf{w}_{kl}[q] s_k[q] = \mathbf{W}_l[q] \mathbf{s}[q], \quad (31)$$

where $\mathbf{w}_{kl}[q] \in \mathbb{C}^M$ is the precoding vector that AP l applies to user k , $\mathbf{s}[q] = [s_1[q] \dots s_K[q]]^\top \in \mathbb{C}^K$ collects the data symbols, and $\mathbf{W}_l[q] = [\mathbf{w}_{l1}[q] \dots \mathbf{w}_{lK}[q]] \in \mathbb{C}^{M \times K}$ is the precoding matrix of AP l . Collecting the per-AP precoders of a given user into $\mathbf{w}_k[q] = [\mathbf{w}_{k1}[q]^\top \dots \mathbf{w}_{kL}[q]^\top]^\top \in \mathbb{C}^{LM}$ yields the network-wide precoding matrix

$$\mathbf{W}[q] = [\mathbf{w}_1[q] \dots \mathbf{w}_K[q]] \in \mathbb{C}^{LM \times K}, \quad (32)$$

whose l -th block of M rows equals $\mathbf{W}_l[q]$, so that the stacked transmit vector is $\mathbf{x}[q] = [\mathbf{x}_1[q]^\top \dots \mathbf{x}_L[q]^\top]^\top = \mathbf{W}[q] \mathbf{s}[q] \in \mathbb{C}^{LM}$.

The signal received by user k on subcarrier q is $y_k[q] = \mathbf{h}_k[q]^H \mathbf{x}[q] + n_k[q]$, which, using (32), separates into the desired term, the multi-user interference, and the noise,

$$y_k[q] = \underbrace{\mathbf{h}_k[q]^H \mathbf{w}_k[q] s_k[q]}_{\text{desired signal}} + \underbrace{\sum_{\substack{i=1 \\ i \neq k}}^K \mathbf{h}_k[q]^H \mathbf{w}_i[q] s_i[q]}_{\text{multi-user interference}} + \underbrace{n_k[q]}_{\text{noise}}, \quad (33)$$

where $n_k[q] \sim \mathcal{CN}(0, \sigma^2)$ is the receiver noise with power σ^2 , independent across users and subcarriers and independent of the channels and data. The scalar $\mathbf{h}_k[q]^H \mathbf{w}_k[q]$ is the effective channel seen by user k : the distributed precoder turns the LM -antenna channel into an equivalent single-antenna link.

Unlike co-located massive MIMO, where a single sum-power budget applies to the whole array, each AP is a separate transmitter and is subject to its own power constraint. Since the data symbols are independent and unit-power, the average power radiated by AP l across the OFDM block is

$$\sum_{q=1}^Q \mathbb{E}(\|\mathbf{x}_l[q]\|^2) = \sum_{q=1}^Q \sum_{k=1}^K \mathbb{E}(\|\mathbf{w}_{kl}[q]\|^2) \leq \rho_{\text{max}}, \quad l = 1, \dots, L, \quad (34)$$

where ρ_{max} is the maximum downlink transmit power per AP, assumed equal for all APs, and the expectation is taken over the data and channel realizations.

It is convenient to split each precoder into a transmit direction and a power control coefficient,

$$\mathbf{w}_k[q] = \sqrt{\rho_k} \frac{\bar{\mathbf{w}}_k[q]}{\sqrt{\mathbb{E}(\|\bar{\mathbf{w}}_k[q]\|^2)}}, \quad (35)$$

where $\bar{\mathbf{w}}_k[q] \in \mathbb{C}^{LM}$ is an arbitrarily scaled vector that fixes the spatial direction of the transmission to user k and $\rho_k \geq 0$ is the downlink power allocated to that user, so that $\mathbb{E}(\|\mathbf{w}_k[q]\|^2) = \rho_k$. This separation isolates the two design

tasks treated below: the choice of the precoding directions $\{\bar{\mathbf{w}}_k[q]\}$ (Section II-A1) and the choice of the power allocation $\{\rho_k\}$ (Section II-A2), the latter being what ties the per-user powers to the per-AP budget in (34).

Treating the multi-user interference in (33) as noise, the effective downlink signal-to-interference-plus-noise ratio (SINR) of user k on subcarrier q is

$$\text{SINR}_k^{\text{dl}}[q] = \frac{|\mathbf{h}_k[q]^H \mathbf{w}_k[q]|^2}{\sum_{\substack{i=1 \\ i \neq k}}^K |\mathbf{h}_k[q]^H \mathbf{w}_i[q]|^2 + \sigma^2}. \quad (36)$$

With the effective channel known at the receiver, an achievable rate for user k on subcarrier q is $\log_2(1 + \text{SINR}_k^{\text{dl}}[q])$, and the delivered downlink sum rate R_{DL} follows by inserting these per-user, per-subcarrier SINRs into the ergodic rate expression of Section I-F ((28)), which in turn drives the encoder and MIMO processing consumption of the power model.

1) *Transmit Precoding Schemes*: The precoding directions can be computed in two fundamentally different ways, mirroring the centralized and distributed uplink architectures of [2]. In *centralized* operation the APs forward their received pilots to the CPU, which estimates the collective channels and designs all directions jointly. Because it sees the whole array, the CPU can make the serving APs cancel one another's interference, giving up to LM spatial degrees of freedom for suppressing the interference to each user. The nearly optimal choice is minimum mean-squared-error (minimum mean squared error (MMSE)) precoding, the downlink dual of MMSE combining, which balances the signal power delivered to a user against the interference it causes; its complexity grows with K , so it is not scalable. Scalable approximations restrict the computation to the users that share serving APs with the target user, giving the P-MMSE and P-RZF precoders, which retain most of the interference-suppression ability at a bounded per-user cost. Removing the channel-estimate regularization altogether yields regularized zero-forcing and, in the limit, the zero forcing (ZF) precoder

$$\bar{\mathbf{W}}[q] = \mathbf{H}[q](\mathbf{H}[q]^H \mathbf{H}[q])^{-1}, \quad (37)$$

$$\mathbf{H}[q] = [\mathbf{h}_1[q] \cdots \mathbf{h}_K[q]] \in \mathbb{C}^{LM \times K}, \quad (38)$$

whose columns $\bar{\mathbf{w}}_k[q]$ null the inter-user interference when the channel state information (CSI) is accurate. Zero-forcing is the scheme adopted in the power model of Section I; regularized zero forcing (RZF), obtained by adding a loading term $\mathbf{H}[q](\mathbf{H}[q]^H \mathbf{H}[q] + \lambda \mathbf{I}_K)^{-1}$, trades some interference leakage for robustness at low SNR. In practice the CPU builds $\mathbf{H}[q]$ from the channel estimates acquired on the uplink pilots, exploiting TDD reciprocity so that no downlink pilots are needed.

In *distributed* operation the CPU only encodes the data, and each AP forms its own directions $\bar{\mathbf{w}}_{kl}[q]$ from its locally estimated channels $\hat{\mathbf{h}}_{kl}[q]$, the MMSE estimates of $\mathbf{h}_{kl}[q]$. An AP can then suppress only the interference it generates itself, with the M spatial degrees of freedom of its own array,

so the coherent cross-AP cancellation of the centralized case is lost and interference must instead be kept small at each AP individually. The local counterpart of MMSE precoding is L-MMSE precoding, which is locally optimal but again not scalable; LP-MMSE precoding restores scalability by summing only over the users an AP serves; and the simplest option is maximum-ratio (maximum ratio (MR)) precoding,

$$\bar{\mathbf{w}}_{kl}[q] = \mathbf{h}_{kl}[q], \quad (39)$$

which maximizes the signal power delivered to the desired user but ignores the interference it causes, with the benefit of admitting a closed-form SINR. Whichever architecture is used, the uplink-downlink duality guarantees that any set of uplink combining vectors achieves the same SINRs when reused, after normalization, as downlink precoding directions [2].

2) *Downlink Power Control*: The directions fix the shape of each user's beam but not how much power it receives; that is set by the power control coefficients $\{\rho_k\}$, which must satisfy the per-AP constraints in (34) while distributing a limited budget across users. Let $\bar{\mathbf{w}}'_{kl}[q] \in \mathbb{C}^M$ be the AP- l block of the normalized direction $\bar{\mathbf{w}}_k[q]/\sqrt{\mathbb{E}(\|\bar{\mathbf{w}}_k[q]\|^2)}$, so that $\mathbb{E}(\|\bar{\mathbf{w}}'_{kl}[q]\|^2)$ is the fraction of user k 's power that AP l radiates. The constraint (34) then reads

$$\sum_{q=1}^Q \sum_{k=1}^K \rho_k \mathbb{E}(\|\bar{\mathbf{w}}'_{kl}[q]\|^2) \leq \rho_{\max}, \quad l = 1, \dots, L, \quad (40)$$

which couples the powers of all users served by a common AP. Unlike the directions, which are recomputed every coherence block, the power allocation depends mainly on the large-scale fading and can be held fixed over many coherence blocks.

The simplest policy gives every user the same power and rescales all precoders by a single factor set by the most heavily loaded AP,

$$\gamma = \sqrt{\frac{\rho_{\max}}{\max_l \tilde{p}_l}}, \quad \tilde{p}_l = \sum_{q=1}^Q \sum_{k=1}^K \mathbb{E}(\|\bar{\mathbf{w}}_{kl}[q]\|^2), \quad (41)$$

so that $\mathbf{w}_{kl}[q] = \gamma \bar{\mathbf{w}}_{kl}[q]$. Because the same scalar multiplies every AP, it preserves the relative amplitudes and phases across APs and hence the interference nulling of (37); the most heavily loaded AP then transmits at exactly $\rho_{\max}$ while the others stay below their budget. A per-AP scalar would load every AP fully but, by scaling each AP differently, would perturb the coherent combining and reintroduce residual multi-user interference, so the common factor is used whenever the interference-free SINR in (36) is assumed.

Equal power is wasteful when users have very different channel gains. A more balanced choice is fractional power control, which sets each user's power from its large-scale fading. Writing $\beta_{kl} = \frac{1}{M} \text{Tr}(\mathbf{R}_{kl}[q])$ for the average channel gain between AP l and user k , a common rule in distributed

operation, where AP l assigns a separate power ρ_{kl} to each user through $\mathbf{w}_{kl}[q] = \sqrt{\rho_{kl}} \bar{\mathbf{w}}_{kl}[q] / \sqrt{\mathbb{E}(\|\bar{\mathbf{w}}_{kl}[q]\|^2)}$, is

$$\rho_{kl} = \rho_{\max} \frac{\beta_{kl}^\kappa}{\sum_{i=1}^K \beta_{il}^\kappa}, \quad l = 1, \dots, L, \quad (42)$$

so that AP l splits its full power among its users according to their gains. The exponent κ tunes the trade-off between throughput ($\kappa > 0$, favoring strong users) and fairness ($\kappa < 0$, protecting weak users), with $\kappa = 1/2$ recovering the heuristic of [2]; the centralized case admits the analogous per-user rule ρ_k normalized against the binding constraint (40). Beyond such heuristics, the powers can be optimized for a network utility such as max-min fairness or sum rate, but these problems do not scale to very large networks, so the statistics-based rules above are the practical default.

REFERENCES

- [1] E. Peschiera, S. Yun, Y. Lee, L. V. der Perre, and F. Rottenberg, “A parametric power model of upper mid-band (fr3) base stations for 6g,” in *ICASSP 2026 - 2026 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2026, pp. 21 476–21 480.
- [2] Ö. T. Demir, E. Björnson, and L. Sanguinetti, “Foundations of User-Centric Cell-Free Massive MIMO,” *Foundations and Trends in Signal Processing*, vol. 14, no. 3-4, pp. 162–472, 2021.